Naveen Raman

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Education

Carnegie Mellon University — Pittsburgh, PA
PhD Machine Learning — August 2023 -
Advisor: Fei Fang

University of Cambridge — Cambridge, England
MPhil Advanced Computer Science — October 2022 - June 2023

University of Maryland — College Park, MD
Bachelor of Science - Computer Science and Math, Magna Cum Laude — August 2018 - May 2022

Experience

Harvard — Cambridge, MA
Visiting Researcher — Summer 2026

Amazon — Palo Alto, CA
Research Intern — Summer 2025

World Resource Institute — Washington DC
Research Intern — Spring 2022

MIT Lincoln Labs — Lexington, MA
Research Intern — Summer 2021

Facebook — Menlo Park, CA
Software Engineering Intern — Summer 2020

Software Engineering REU @ Carnegie Mellon — Pittsburgh, PA
Research Intern — Summer 2019

Publications

Refereed Conference and Journal Publications

[C11] "Large Language Models in Peer-Run Community Behavioral Health Services: Understanding Peer Specialists and Service Users' Perspectives on Opportunities, Risks, and Mitigation Strategies," Megan Chai*, Cindy Peng*, Gao Mo, **Naveen Raman**, Shannon Pagdon, Fei Fang, Nev Jones, Peggy Swarbrick, Hong Shen. *Conference on Human Factors in Computing Systems* (CHI) 2026
Best Paper Honorable Mention

[C10] "PeerCoPilot: A Language Model-Powered Assistant for Behavioral Health Organizations," Gao Mo*, **Naveen Raman***, Megan Chai, Cindy Peng, Shannon Pagdon, Nev Jones, Hong Shen, Peggy Swarbrick, Fei Fang. *Innovative Applications of Artificial Intelligence* (IAAI) 2026

[C9] "RescueLens: LLM-Powered Triage and Action on Volunteer Feedback for Food Rescue," **Naveen Raman**, Jingwu Tang, Zhiyu Chen, Zheyuan Ryan Shi, Sean Hudson, Ameesh Kapoor, Fei Fang. *Innovative Applications of Artificial Intelligence* (IAAI) 2026
Deployed Application Award

[J1] "Do Concept Bottleneck Models Obey Locality?" **Naveen Raman**, Mateo Espinosa, Juyeon Heo, Mateja Jamnik. *Transactions on Machine Learning Research* (TMLR) 2025

[C8] "Data-driven Design of Randomized Control Trials with Guaranteed Treatment Effects," Santiago Cortes-Gomez, **Naveen Raman**, Aarti Singh, Bryan Wilder. *International Conference on Machine Learning* (ICML) 2025

[C7] "Assortment Optimization for Patient-Provider Matching," **Naveen Raman**, Holly Wiberg. *Machine Learning for Health* (ML4H) 2024

[C6] "Global Rewards in Restless Multi-Armed Bandits," **Naveen Raman**, Ryan Shi, Fei Fang. *Conference on Neural Information Processing Systems* (NeurIPS) 2024

[C5] "Understanding Inter-Concept Relationships in Concept-Based Models," **Naveen Raman**, Mateo Espinosa Zarlenga, Mateja Jamnik. *International Conference on Machine Learning* (ICML) 2024

[C4] "Human Uncertainty in Concept-Based AI Systems," Katherine M Collins, Matthew Barker, Mateo Espinosa Zarlenga, **Naveen Raman**, Umang Bhatt, Mateja Jamnik, Ilia Sucholutsky, Adrian Weller, Krishnamurthy Dvijotham. *AAAI/ACM Conference on AI, Ethics, and Society* (AIES) 2023

[C3] "Data-Driven Methods for Balancing Fairness and Efficiency in Ride-Pooling," **Naveen Raman**, Sanket Shah, John Dickerson. *International Joint Conference on Artificial Intelligence* (IJCAI) 2021

[C2] "Stress and burnout in open source: Toward finding, understanding, and mitigating unhealthy interactions," **Naveen Raman**, Minxuan Cao, Yulia Tsvetkov, Christian Kästner, Bogdan Vasilescu. *International Conference on Software Engineering New Ideas and Emerging Results* (ICSE NIER) 2020

[C1] "A Muffin-Theorem Generator," Guangqi Cui*, John Dickerson*, Naveen Durvasula*, William Gasarch*, Erik Metz*, Jacob Prinz*, **Naveen Raman***, Daniel Smolyak*, Sung Hyun Yoo*. *International Conference on Fun with Algorithms* (FUN) 2018

Refereed Workshop Papers and Extended Abstracts

[W4] Ariana Tang, **Naveen Raman**, Fei Fang, Ryan Shi. Contextual Budget Allocation for Food Rescue Volunteer Engagement. *MLxOR Workshop at NeurIPS* 2025

[W3] **Naveen Raman**, Michael Yee. Improving Learning-to-Defer Algorithms Through Fine-Tuning. *Workshop on Human and Machine Decisions at NeurIPS* 2021

[W2] Andrew Mao*, **Naveen Raman***, Matthew Shu, Eric Li, Franklin Yang, Jordan Boyd-Graber. Eliciting Bias in Question Answering Models through Ambiguity. *Machine Reading for Question Answering at EMNLP* 2021

[W1] **Naveen Raman**, Pedro Rodriguez, Jordan Boyd-Graber. What more can Entity Linking do for Question Answering? *Human And Machine in-the-Loop Evaluation and Learning Strategies at NeurIPS* 2020

Preprints and Working Papers

[P6] **Naveen Raman**, Fei Fang, Yiling Chen. Strategic Competition and Convergence in LLM Leaderboards.

[P5] **Naveen Raman**, Holly Wiberg. Matching with Choice: Menu Design for Patient-Provider Assignment.

[P4] Yanru Guan, **Naveen Raman**, Fei Fang. Human Decision-Making with AI Assistance under Correlated Features.

[P3] **Naveen Raman***, Santiago Cortes-Gomez*, Mateo Dulce Rubio*, Fei Fang, Bryan Wilder. Healthcare LLM Benchmarks Are Only as Good as Their Explicit Assumptions.

[P2] **Naveen Raman**, Stephanie Milani, Fei Fang. Selecting Decision-Relevant Concepts in Reinforcement Learning.

[P1] **Naveen Raman**, Ryan Shi, Sean Hudson, Ameesh Kapoor, Fei Fang. An Extended Study on Restless Multi-Armed Bandits with Global Rewards.

Honors and Awards

K&L Gates Presidential Fellowship	2026
CHI Best Paper Honorable Mention	2026
IAAI Deployed Application Award	2026
NeurIPS Top Reviewer	2024
University of Maryland Undergraduate Researcher of the Year	2022
National Science Foundation Graduate Research Fellowship (GRFP)	2022
Churchill Scholarship (fully funded Cambridge MPhil)	2022
CRA Undergraduate Researcher of the Year Finalist	2021
Phillip Merrill Presidential Scholar	2021
Barry Goldwater Scholarship	2021
Brendan Iribe Scholarship (full tuition scholarship)	2020
President's Scholarship (4-year scholarship)	2018

Professional Service

Conference Reviewing

Main Conference: NeurIPS (2022,2023, Top Reviewer 2024, 2025), AAAI (2025), ICML (2024,2025, 2026 Silver), AAMAS (2024), COLING (2022), AISTATS (2025,2026), EAAMO (2025,2026), CHIL (2025), ICLR (2026)
Workshop: ICML Actionable Interpretability (2025), AAAI Deployed AI (2026), ICML CompLearn (2026), NeurIPS Judge (2026), NeurIPS InterpScience (2026)

Organizing

CMU ML-AI Seminar *Organizer* — August 2026 - Present
CMU Econ-CS Seminar *Organizer* — January 2026 - Present
NeurIPS Education Program *Assistant Chair* — December 2025
EAAMO *Social Chair* — November 2025
MLC Africa *Volunteer Assistant* — October 2023 - October 2025
Machine Learning for Health *Roundtable Chair* — December 2024

Department Service

Machine Learning Blog *Blog Head Editor* — July 2025 - Present
Machine Learning Department *PhD Admissions Committee* — November 2025 - February 2026
Machine Learning Department *Masters Admissions Committee* — November 2024 - May 2025
Machine Learning Blog *Blog Editor* — August 2024 - June 2025
Machine Learning Department *Social Committee* — October 2023 - May 2024

Mentorship

School of Computer Science *Graduate Application Support Program* — October 2023 - Present
School of Computer Science *AI Mentorship Program* — October 2023 - Present
School of Computer Science *Tech Nights* — October 2023 - December 2023
Student Community Action *Big Sibling* — October 2022- May 2023
Maryland Mentors Program *Math and Reading Tutor* — September 2020-May 2022
College Park Academy *Academic Volunteer* — September 2018-May 2021
College Mentors for Kids *Mentor* — January 2021 - May 2021

Talks

Healthcare LLM Benchmarks and Assumptions
Deep Learning Interpretability Seminar

Selecting Decision-Relevant Concepts in Reinforcement Learning
Stanford AIMS Seminar

RescueLens: LLM-Powered Triage and Action on Volunteer Feedback for Food Rescue
Food Rescue Conference
IAAI'26

Assortment Optimization for Patient-Provider Matching
Marketplace Innovations Workshop
Udell Group at Stanford University
Harvard Econ-CS Group
INFORMS Annual Meeting
Rutgers Econ-CS Group
AAAI Combining AI and OR/MS for Better Trustworthy Decision Making
Manufacturing and Service Operations Management Annual Conference

PeerCoPilot: A Language-Model Assistant for Behavioral Health Organizations
Carnegie Mellon FEAT Reading Group
AI-SDM Workshop on AI for Social Impact
Amazon Applied AI Seminar
University of Maryland CLIP Colloquium
Columbia HCI Seminar Series
IAAI'26

Global Rewards in Restless Multi-Armed Bandits
Food Rescue Conference

University of Maryland Multi-Agent RL Group
Carnegie Mellon Econ-CS Reading Group
Multi-Agent Learning Seminar
EAAMO Urban Data Science Working Group

Teaching

Teaching Assistant
Advanced Introduction to Machine Learning (10-715), CMU Fall 2025
Organization of Programming Languages (CMSC330), UMD Fall 2019 - Spring 2022
Introduction to Programming via the Web (CMSC122), UMD Spring 2019

Course Facilitator
Algorithms for Interviews (CMSC389O), UMD Spring 2020 - Spring 2022